Homework Assignment 1: Mapping Environmental Injustices

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Import Data / Load Packages

```
library(tidyverse)
library(sf)
library(here)

# install packages to map the data
# install.packages("stars")
# install.packages("tmap")

# read in geodatabase of EJScreen data at the Census Block Group level
ejscreen <- sf::st_read(here::here("data", "ejscreen","EJSCREEN_2023_BG_StatePct_with_AS_CNM
```

```
Reading layer `EJSCREEN_StatePctiles_with_AS_CNMI_GU_VI' from data source
  `C:\Users\aaron\Documents\Git\eds223-assignment-1\data\ejscreen\EJSCREEN_2023_BG_StatePct_w
  using driver `OpenFileGDB'
Simple feature collection with 243021 features and 223 fields
Geometry type: MULTIPOLYGON
Dimension:     XY
Bounding box:  xmin: -19951910 ymin: -1617130 xmax: 16259830 ymax: 11554350
Projected CRS: WGS 84 / Pseudo-Mercator
```

```
# filter to a state you are interested in
california <- ejscreen %>%
  dplyr::filter(ST_ABBREV == "CA")

# filter to a county you are interested in
```

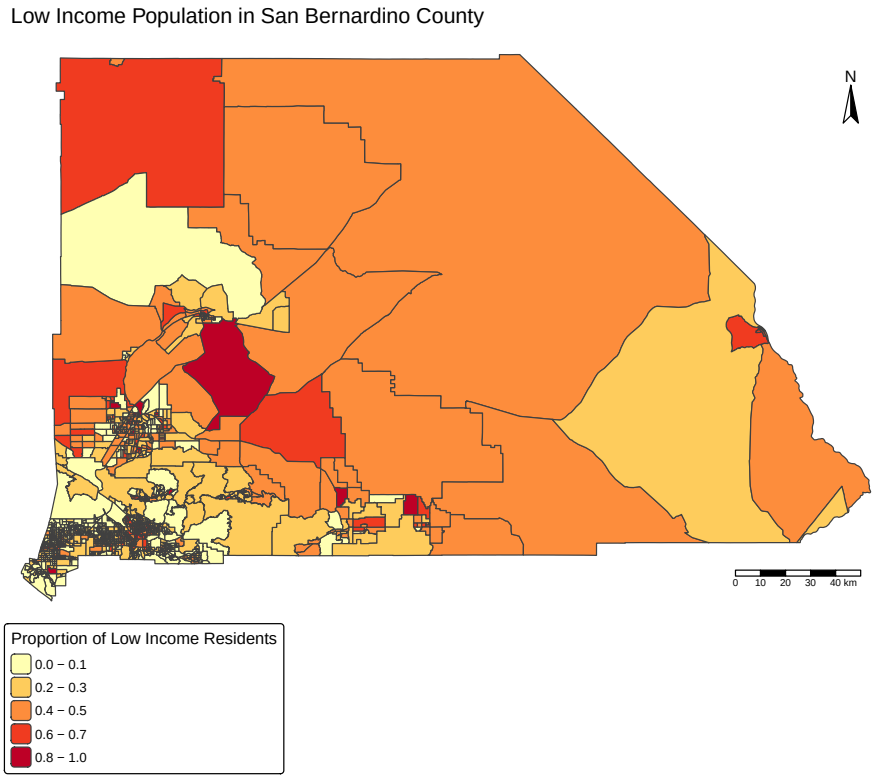
```r
san_bernardino <- ejscreen %>%
  dplyr::filter(CNTY_NAME %in% c("San Bernardino County"))

# find the average values for all variables within counties
california_counties <- aggregate(california, by = list(california$CNTY_NAME), FUN = mean)
```

Visualize Data

```r
library(sf)
library(stars)
library(tmap)
library(here)

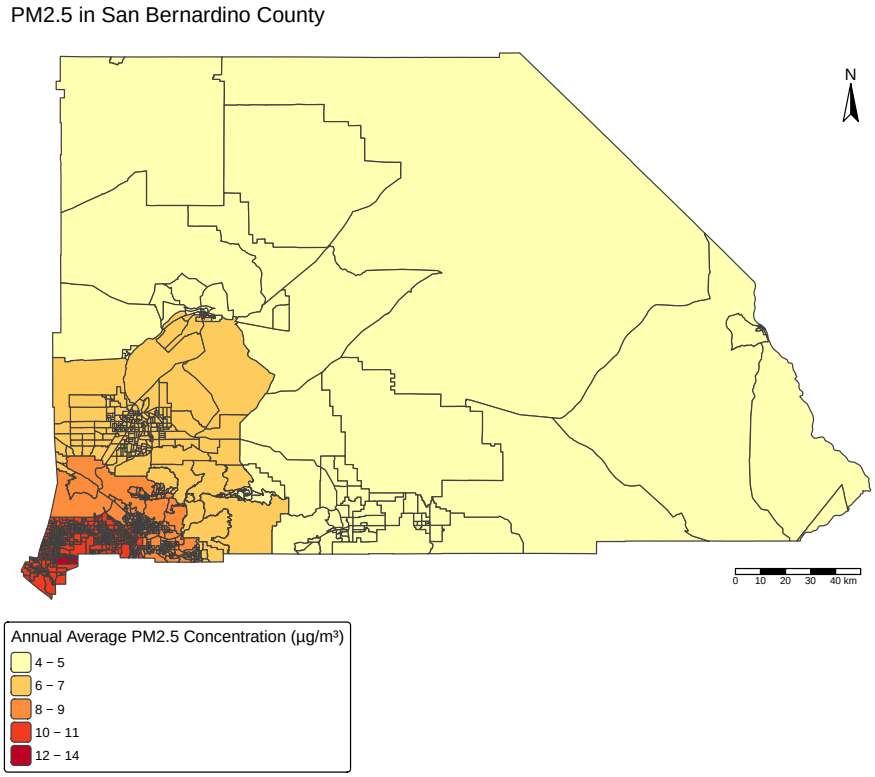
# map the low income proportion in the scale of San Bernardino county
tm_shape(san_bernardino) +
  tm_polygons('LOWINCPCT',
              fill.legend = tm_legend(title = 'Proportion of Low Income Residents'),
              fill.scale = tm_scale(values = 'YlOrRd')
  ) +
  tm_title('Low Income Population in San Bernardino County') +
  tm_compass(position = c('top', 'right')) +
  tm_scalebar(
    position = c('bottom', 'right'),
    size = 1
  ) +
  tm_layout(
    frame = FALSE,
    title.size = 1.1,
    legend.text.size = 0.7,
    legend.title.size = 0.8
  )
```




```
# map the PM2.5 in the scale of San Bernardino county
tm_shape(san_bernardino) +
  tm_polygons('PM25', #PTSDF
              fill.legend = tm_legend(title = 'Annual Average PM2.5 Concentration (µg/m³)'),
              fill.scale = tm_scale(values = 'YlOrRd')

  ) +
  tm_title('PM2.5 in San Bernardino County') +
  tm_compass(position = c('top', 'right')) +
  tm_scalebar(
    position = c('bottom', 'right'),
    size = 1
  ) +
  tm_layout(
    frame = FALSE,
    title.size = 1.1,
    legend.text.size = 0.7,
    legend.title.size = 0.8
```

```
)
```



Map Interpretation

The PM2.5 map shows that air pollution is generally higher in the southwest part of the county where it is densely populated thus showing that more transportation and urban activities are apparent. In constrast the low income map demonstrates that the distribution of the proportion of low income residents are more spread out across the county. In the southwest region, there are both relatively high PM2.5 concentrations and low income residents which indicates that there is potential environmental injustice. Despite this, the relationship between the two factors is inconsistent throughout the entirety of San Bernardino county. In the western and eastern parts of the map, there are more rural communities which contributes to the higher proportion of low income residents who are not necessarily experiencing higher concentrations of PM2.5.

Data Citation

Data	Citation	Link
PM2.5	U.S. EPA. 2023. EJScreen Version 2.2.	EPA EJScreen
Low Income	U.S. EPA. 2023. EJScreen Version 2.2.	EPA EJScreen

Data sources used in the San Bernardino County environmental justice analysis. The original website no longer exists but the link provided is an alternative.